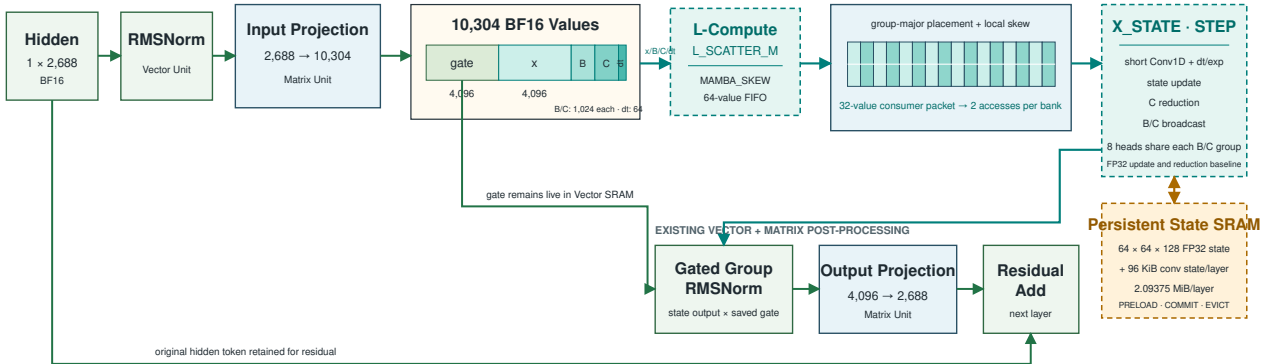


Nemotron 3 Mamba-2 Decode: Producer-Consumer Co-Layout

One token · hidden size 2,688 · 64 heads · head dimension 64 · state dimension 128 · 8 B/C groups

EXISTING PLENA FRONT END



Projection-buffer service ROW_MAJOR **53,176 cycles** → **MAMBA_SKEW** **14,904 cycles** **3.568× local** 236,992 values exact bank stalls: 0 HBM repack: 0 B

The reported speedup is the verified Matrix-SRAM consumer service time for 23 Mamba layers (B1 decode), not end-to-end model speedup.